Study Unit 6 – Files

V2 - Files and Exceptions

1. Validating Number Input

Write a program that asks the user to enter a number. Use a try and except to catch any error if the user does not enter a number. If the input is valid, print the number. If not, print an error message.

2. Handling Missing Files

Write a program that tries to open a file named data.txt and read its contents. If the file does not exist, catch the error and print an error message.

3. Writing Employee Records Safely

Write a program that creates a file named Employees.txt and writes the following names in the file:

- John Doe
- Jane Dunning
- James Potter

Add error handling in case the file cannot be created or written to.

4. Safe Division with Error Handling

Write a program that asks the user to enter two numbers. Divide the first number by the second. Use try and except to handle invalid input and also division by zero.

5. Saving Trip Names to a File

Write a program that asks the user to enter three names (one at a time). Save each name on a new line in a file called Trip.txt. Use try and except to handle any errors that may happen during writing.

6. Reading and Summing Sales Values

Create a program that opens a file named sales.txt, reads three lines from it, and tries to add them as numbers. If any line is not a number, catch the error and skip it. Show the total at the end.

7. Reading a File with Confirmation

Write a program that asks the user to enter a file name. Try to open and read the file. If the file is found, print its contents and display a success message using an `else` clause. If the file is not found, print an error message.

8. Trip Contributions with Validation

Write a program that asks the user for three names and the amount each person wants to contribute to a trip. Save each name and amount to a file called `Trip.txt`. Use error handling in case the user enters something that is not a number.

9. Reading with a Final Message

Write a program that tries to open and read from a file called `Employees.txt`. If the file is not found, print an error message. Use a `finally` clause to print a message that always runs at the end.